

Hands-on with SPARK: Multiple Flows & the Flow Control Panel

Run two independent flows - a text summariser and an image describer - on one canvas, controlling each from the Flow Control Panel.

Activity is designed by Su Nguyen; Detailed guide is drafted by Claude Cowork.

Before you start

Open SPARK in your browser. You need an API key set in Settings (gear, top-right). The activity uses gemini-2.5-flash. The image used in the second flow (arsenal-football.jpg) is the same one from Week 6 (Week 6 > data > images).

Tool	SPARK - Studio for Prototyping Agent Reasoning & Knowledge
Example	Dual Flow Test
Flow 1	My Text -> Summarizer -> Show Summary
Flow 2	My Image -> Describer -> Show Description
Key idea	Several flows can live on one canvas; the Flow Control Panel lets you define and run each one separately

Step 1 - Open the Dual Flow Test example

Until now each canvas held a single flow. This example shows that a canvas can hold several independent flows at once - useful when you are prototyping a few small tools side by side.

1. On the Gallery tab, find the Dual Flow Test card ("Contains two independent flows on one canvas to test the Flow Control panel").
2. Click Load Example (Confirm if asked to clear the canvas).
3. The Canvas shows two separate chains: a text flow (My Text -> Summarizer -> Show Summary) on top, and an image flow (My Image -> Describer -> Show Description) below.

Both LLM nodes (Summarizer and Describer) use the same model. The example loads with a GPT model (gpt-5.4-nano); you can switch either node to Google Gemini in its Properties if you prefer - the activity slide shows gemini-2.5-flash. Please check how to configure AI providers and models from previous SPARK activities.

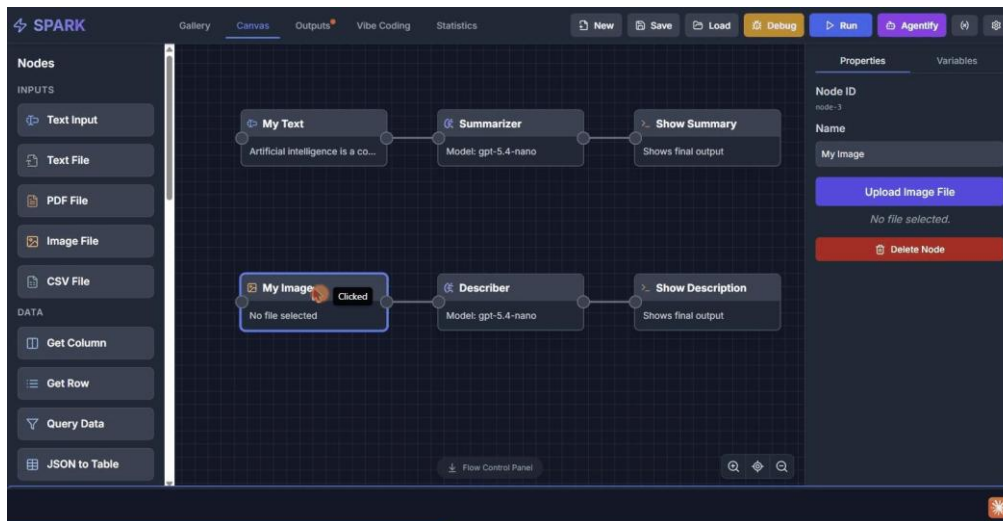


Figure 1 - Two independent flows on one canvas. Click the My Image node and use Upload Image File to add arsenal-football.jpg to the second flow.

Step 2 - Open the Flow Control Panel

With more than one flow on the canvas, the Run button is ambiguous. The Flow Control Panel is where you tell SPARK which nodes belong to which flow, and run each flow on its own.

4. Move your mouse to the bottom of the canvas and click the Flow Control Panel tab to open it.
5. Use the dropdown at the top of the panel to choose a flow - start with Text Summarizer Flow.
6. Each flow has a Flow Name, a Context/Goal, a set of Triggers (Input Nodes), and Output Collectors (Sink Nodes). For the text flow, the trigger is My Text and the output collector is Show Summary.

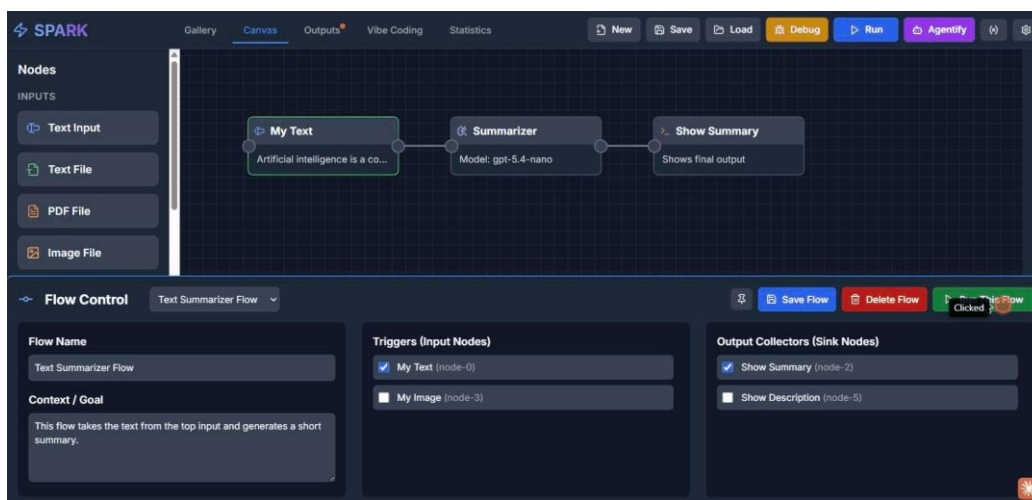


Figure 2 - The Flow Control Panel: the Text Summarizer Flow is defined by its trigger (My Text) and output (Show Summary).

Tip: the checkboxes define which input/output nodes belong to the selected flow. This is how SPARK keeps two flows on one canvas separate.

Step 3 - Run the text flow

With Text Summarizer Flow selected, click Run This Flow (top-right of the panel). Only that flow runs - the top chain lights up, the bottom one stays idle - and the summary appears in an Agent Output dialog.

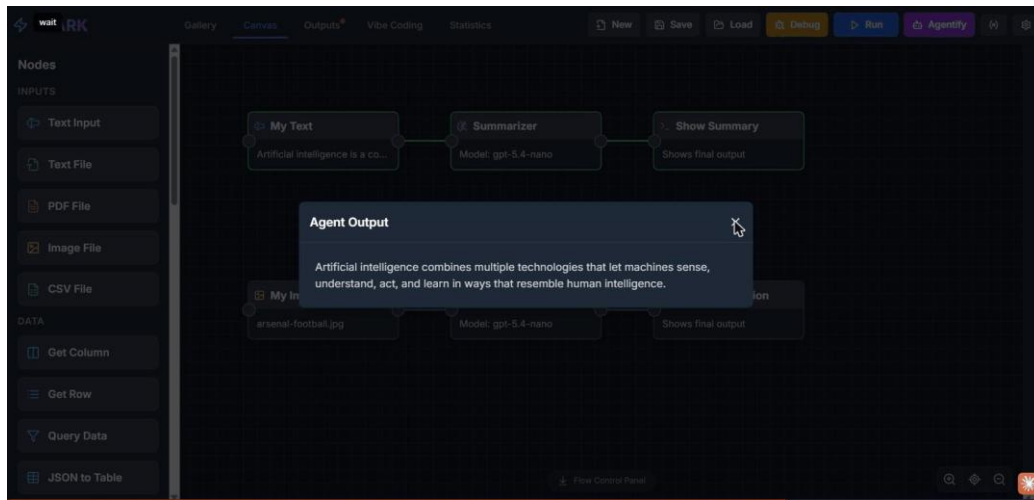


Figure 3 - Run This Flow on the text flow only: the Summarizer condenses the input text into a single clear sentence.

Step 4 - Switch flows and run the image flow

Now switch the Flow Control dropdown to Image Descriptor Flow. Its trigger is My Image and its output is Show Description. Click Run This Flow again.

This time the Descriptor looks at arsenal-football.jpg (upload the image if you have not done so) and returns a detailed description of the scene - the goalmouth, the players jumping for the ball, the crowded stadium. The same canvas, a completely different flow, run on its own.

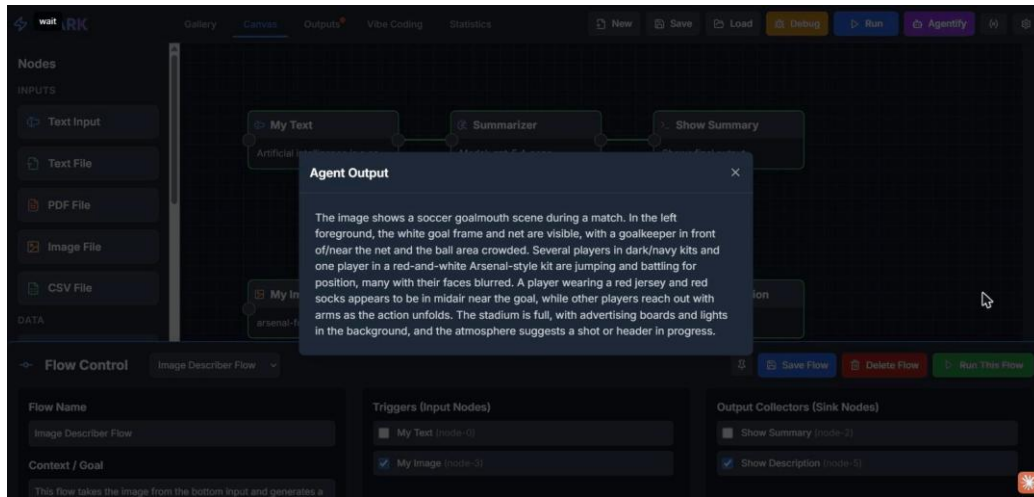


Figure 4 - Switch to the Image Describer Flow and Run This Flow: the second flow produces a description of the image, independent of the text flow.

Good to know

- One canvas can hold many independent flows - handy for prototyping several small tools together.
- The Flow Control Panel is what makes this work: define each flow's input (Trigger) and output (Output Collector), then Run This Flow to run just that one.
- The top Run button runs the whole canvas; Run This Flow (in the panel) runs only the selected flow.
- Both LLM nodes share a model setting - switch the AI Provider per node if Gemini is busy or you want to compare models.
- These named flows become the building blocks for AI agents later in Week 7 - each flow can be exposed to an agent as a tool.